Continuous Optimization for Airline Hub-and-Spoke Network Design

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Abstract

We study the design of airline hub-and-spoke networks through a continuous optimization framework that jointly models network deployment, passenger choice, and market profitability. The lower-level model combines operating costs, sparse deployment penalties, price-weighted passenger utility terms, and entropy regularizers that recover a logit demand split between the new airline and the outside alternative. This yields a continuous approximation of hub activation and route opening decisions while preserving the economic structure of the airline design problem.

On top of this model, we introduce a bilevel formulation in which market weights are learned rather than fixed. The upper level maximizes the real objective of the airline, namely profit, by adjusting market-specific hyperparameters that scale the lower-level utility and entropy terms. This allows the model to emphasize the origin-destination markets that are most relevant for profitability, instead of relying on exogenous weights chosen *a priori*.

The main difficulty is the simultaneous presence of hierarchical optimization and fixed deployment costs. To address it, we combine a reweighted ℓ_1 approximation of fixed costs with a penalty-based bilevel strategy. The resulting algorithm alternates an outer sparsity loop, which updates the reweighting coefficients associated with capacities, and an inner penalty loop, which alternates between a penalized bilevel solve and a lower-level solve to update the market hyperparameters. The final methodology provides a practical route to solving airline network design problems that are beyond the reach of exact mixed-integer bilevel formulations, while maintaining an explicit link between passenger behavior, network topology, and profit.

1 Introduction

Airline hub-and-spoke design requires deciding which airports should operate as transfer centers, which direct connections should be offered, how much capacity should be assigned to each airport and link, and how origin-destination demand should be routed through the resulting network. These decisions are tightly coupled. Modifying the hub structure changes the set of feasible itineraries, which changes passenger utility and market capture, which in turn alters the profit associated with each route and the value of investing in additional capacity.

This interaction makes exact formulations difficult to solve at realistic scales. Mixed-integer models can represent route opening, hub activation, and fixed costs naturally, but they become computationally challenging once they are combined with multicommodity flows and demand-choice models. In competitive airline markets, the difficulty is even greater because the new airline must be designed while accounting for outside alternatives and for the fact that different markets contribute differently to profit.

This paper adopts a continuous optimization viewpoint. Instead of modeling build-or-not-build decisions with binary variables, we use sparse capacity variables and reweighted penalties to emulate

fixed costs. Instead of imposing logit demand capture through non-convex constraints, we recover it through entropy terms in the objective. The result is a single-level continuous model that remains rich enough to represent the economic structure of airline hub-and-spoke design.

Our main contribution is to extend that formulation to a bilevel setting. The lower level is a continuous airline network design model with price-weighted utility and entropy terms, reweighted deployment costs, and hub-capacity restrictions. The upper level maximizes profit by learning market-specific hyperparameters that determine how strongly each origin-destination market is represented inside the lower-level passenger terms. In this sense, the bilevel layer calibrates the lower-level model so that the network induced by continuous optimization is better aligned with the true business objective of the airline.

To solve the resulting problem, we use a penalty-based strategy. The method combines an outer loop that updates the reweighted approximation of fixed costs with an inner loop that alternates between a penalized bilevel solve and a lower-level solve, followed by projected updates of the market hyperparameters. This design keeps the computational advantages of the continuous model while introducing a principled mechanism to steer the network toward higher-profit configurations.

The paper is organized as follows. Section 2 reviews the main modeling directions that motivate the approach. Section 3 introduces the airline hub-and-spoke setting and the optimization variables. Section 4 presents the single-level lower-level model. Section 5 formulates the bilevel extension and its penalty-based resolution strategy. Section 6 discusses the computational implementation, and Section 7 concludes.

2 State of the Art

Classical airline hub-and-spoke design models are usually formulated as mixed-integer network design problems. Binary variables decide whether an airport is activated, whether it operates as a hub, and whether a direct connection is offered, while continuous variables describe traffic assignment and frequencies. These formulations are expressive and remain the standard modeling choice when the problem size is moderate, but they become extremely demanding when fixed costs, multicommodity flows, demand choice, and realistic operational constraints are included simultaneously.

A second relevant line of work concerns demand modeling. In many network design models, passenger demand is treated either through deterministic all-or-nothing assignment or through logit-based market shares. The latter is more realistic in competitive settings, but it often leads to non-convex constraints or to piecewise-linear mixed-integer approximations. This substantially increases the computational burden and makes the resulting formulations difficult to scale.

These limitations have motivated continuous alternatives in which discrete activation decisions are replaced by sparse capacity variables and logit behavior is recovered through convex regularization. In that family of approaches, two ideas are particularly useful. First, reweighted ℓ_1 penalties derived from logarithmic sparsity surrogates can emulate fixed activation costs while preserving a continuous optimization structure. Second, entropy terms in the objective can induce logit demand shares through first-order optimality conditions, avoiding the explicit enforcement of non-convex logit constraints.

The present work combines those ideas in the airline hub-and-spoke setting and extends them with a bilevel layer. The lower level remains a continuous network design model, but the market weights entering the passenger terms are now learned rather than fixed. For the resulting constrained bilevel problem, our resolution strategy follows the recent penalty-method viewpoint advocated in the bilevel optimization literature, where upper- and lower-level objectives are coupled

through penalized single-level surrogates that can be treated by repeated first-order or gradient-based updates.

3 Problem Setting

We consider a set of candidate airports indexed by $i, j \in \mathcal{S}$ and a set of origin-destination markets indexed by $(o, d) \in \mathcal{K}$, with $o \neq d$. For each market, the new airline competes against an outside alternative representing incumbent competitors already operating in the market. The network to be designed follows a hub-and-spoke structure: some airports only support local traffic, while others are assigned additional transfer capacity and operate as hubs.

The main continuous design variables are:

- airport-capacity variables s_i and hub-capacity variables sh_i, corresponding to the non-hub and hub components of installed airport capacity;
- link-capacity or frequency variables a_{ij} for each potential flight leg (i, j);
- market-share variables f_{od} and f_{od}^{ext}, representing the proportions of market (o, d) captured by the new airline and by the outside alternative, respectively;
- routing variables f_{ij}^{od}, which determine how the demand captured by the airline is assigned over the arcs of the designed network.

The state data include market demand, ticket prices, travel times, operating cost coefficients, airport deployment costs, hub costs, and the set of candidate direct connections. Given these quantities, the optimization determines the network topology, the installed capacities, the market shares, and the route assignment of the captured demand.

Passenger behavior is modeled through a two-alternative choice between the new airline and the outside option. Rather than imposing the corresponding logit split through an explicit non-linear constraint, the model introduces entropy and utility terms in the objective. The optimality conditions of the lower-level problem then yield the market split endogenously.

The hub-and-spoke structure is enforced through dedicated capacity constraints. Airport capacity must sustain departures and arrivals, hub capacity must sustain connecting traffic, and direct service between two airports is allowed only when the corresponding endpoints have sufficient hub capacity. In addition, market-specific profitability constraints are introduced in the airline model so that the network only captures traffic on markets whose revenue covers the associated operating cost.

Finally, deployment decisions remain continuous but are intended to behave sparsely. Positive values of s_i, sh_i, or a_{ij} indicate that the corresponding resource is activated, while zero values indicate that it is not used. This is the mechanism through which the continuous model emulates the discrete build-or-not-build logic of the original mixed-integer problem.

4 Single-Level Multiobjective Formulation

This section presents the lower-level problem used inside the bilevel scheme. Let

$$x := \left(s, sh, a, f, f^{\text{ext}}, f_{ij}^{od}\right) \tag{1}$$

collect the airport-capacity, hub-capacity, link-capacity, market-share, and route-flow variables. The lower level is a continuous airline network design model in which fixed deployment costs are

approximated by reweighted penalties and the passenger split between the new airline and the outside option is induced through entropy regularization.

To separate the market-wise bilevel hyperparameters from the scalar penalty coefficient used for deployment costs, we denote the former by $\widehat{\alpha}^{od}$ and $\widehat{\beta}^{od}$. The lower-level market weight for market (o, d) is

$$W^{od} := \widehat{\alpha}^{od}\big(p^{od}q^{od}\big)^{\widehat{\beta}^{od}}, \tag{2}$$

where q^{od} and p^{od} denote demand and price. In the code, these quantities correspond to `alfa_od` and `beta_od`.

The lower-level objective is written as

$$\min_{x}\ \Phi_{\mathrm{LL}}(x; \widehat{\alpha}, \widehat{\beta}) := OP(a) + DEP_{\beta}(s, sh, a) + U_{\mathrm{self}}(x) + U_{\mathrm{ext}}(x) + H_{\mathrm{self}}(x) + H_{\mathrm{ext}}(x), \tag{3}$$

where the different terms have the following meaning.

Operating costs. The direct operating cost is modeled linearly as

$$OP(a) = \sum_{(i,j)\in\mathcal{A}} c_{o_{ij}} a_{ij}. \tag{4}$$

Reweighted deployment costs. The deployment term penalizes airport activation, hub activation, and, at the formulation level, link deployment. Its generic structure is

$$\begin{aligned} DEP_{\beta}(s, sh, a) = \beta\Bigg[&\sum_{(i,j)\in\mathcal{A}} \frac{c_{A_{ij}} a_{ij}}{a_{ij}^{\mathrm{prev}} + \varepsilon_A} + \sum_{i\in\mathcal{S}} \frac{c_{S_i}(s_i + sh_i)}{s_i^{\mathrm{prev}} + sh_i^{\mathrm{prev}} + \varepsilon_S} \\ &+ \sum_{i\in\mathcal{S}} \lambda \frac{c_{H_i} sh_i}{sh_i^{\mathrm{prev}} + \varepsilon_S} + \sum_{(i,j)\in\mathcal{A}} \bar{c}_{A_{ij}} a_{ij} + \sum_{i\in\mathcal{S}} \bar{c}_{S_i}(s_i + sh_i)\Bigg]. \end{aligned} \tag{5}$$

Here β is the scalar penalty coefficient associated with the deployment budget and λ modulates the cost of activating hubs. In the current implementation, the reweighted fixed-cost approximation is concentrated on airport and hub capacities, which are the variables that dominate the fixed-cost nonsmoothness in the airline case.

Utility terms. The utility contribution of passengers traveling on the new airline is written as

$$U_{\mathrm{self}}(x) = -\sum_{(o,d)\in\mathcal{K}} W^{od} p^{od} \left[\omega_p p^{od} f^{od} \omega_t \sum_{(i,j)\in\mathcal{A}} \big(t_{ij} + t_{ij,\mathrm{trans}}^{od}\big) f_{ij}^{od} \right], \tag{6}$$

whereas the competing outside option contributes

$$U_{\mathrm{ext}}(x) = -\sum_{(o,d)\in\mathcal{K}} W^{od} p^{od} \big(\omega_t t_{\mathrm{ext}}^{od} + \omega_p p_{\mathrm{ext}}^{od}\big) f_{\mathrm{ext}}^{od}. \tag{7}$$

The multiplication by p^{od} is essential in the airline setting because the lower level must distinguish not only between large-demand markets, but also between markets that generate different revenue levels.

Entropy terms and logit shares. The logit split is induced by the entropy terms

$$H_{\text{self}}(x) = \sum_{(o,d)\in\mathcal{K}} W^{od} p^{od} f^{od} \big(\log(f^{od}) - 1\big), \tag{8}$$

$$H_{\text{ext}}(x) = \sum_{(o,d)\in\mathcal{K}} W^{od} p^{od} f^{od}_{\text{ext}} \left(\log\left(\frac{f^{od}_{\text{ext}}}{n_{\text{airlines}}}\right) - 1\right). \tag{9}$$

These terms replace the explicit nonlinear logit constraints that would appear in a classical mixed-integer model and produce the market split through the optimality conditions of the continuous problem.

Constraints. The lower-level objective is minimized subject to:

- link-capacity constraints relating traffic and frequencies;
- airport-capacity constraints for arrivals and departures;
- hub-capacity constraints for transfer traffic;
- direct-flight and connectivity constraints induced by the hub-and-spoke structure;
- symmetry constraints on operated links;
- flow-conservation constraints for every market;
- market split constraints $f^{od} + f^{od}_{\text{ext}} = 1$;
- profitability constraints ensuring that route revenue dominates operating cost;
- nonnegativity and candidate-link restrictions.

This lower-level problem is the model that the bilevel scheme repeatedly solves and reweights. The role of the upper level is not to change the feasible set, but to learn the coefficients $\widehat{\alpha}^{od}$ and $\widehat{\beta}^{od}$ that determine how the lower-level passenger terms value each market.

5 Bilevel Weight-Adjustment Framework

The bilevel model treats the market-wise coefficients $\widehat{\alpha}^{od}$ and $\widehat{\beta}^{od}$ as upper-level decision variables. Their role is to determine how strongly each origin-destination market contributes to the lower-level passenger and entropy terms. The upper-level objective is the actual profit of the airline,

$$\Pi(x) = \sum_{(o,d)\in\mathcal{K}} p^{od} q^{od} f^{od} - \sum_{(i,j)\in\mathcal{A}} c_{o_{ij}} a_{ij}, \tag{10}$$

while the lower-level response is the solution of the continuous network design problem introduced in the previous section.

Using the lower-level variable y to emphasize the nested structure, the bilevel problem can be written as

$$\max_{\widehat{\alpha},\widehat{\beta},x} \quad \Pi(x) \tag{11}$$

$$\text{s.t.} \quad x \in \arg\min_{y} \Phi_{\text{LL}}(y; \widehat{\alpha}, \widehat{\beta}), \tag{12}$$

$$\underline{\alpha} \le \widehat{\alpha}^{od} \le \overline{\alpha}, \quad \underline{\beta} \le \widehat{\beta}^{od} \le \overline{\beta} \qquad \forall (o,d) \in \mathcal{K}. \tag{13}$$

The box constraints reflect the projected updates used in the implementation and keep the market hyperparameters in a numerically meaningful range.

Penalty reformulation

Solving the bilevel problem directly is difficult because the lower-level solution set depends implicitly on the hyperparameters. Following the penalty-method viewpoint of recent bilevel optimization algorithms, we work instead with a penalized surrogate in which upper- and lower-level objectives are coupled in a single objective. For a penalty parameter $\gamma > 0$, we consider

$$\mathcal{P}_\gamma(x; \widehat{\alpha}, \widehat{\beta}) = -\Pi(x) \; + \; \gamma \, \Phi_{\mathrm{LL}}(x; \widehat{\alpha}, \widehat{\beta}) \; + \; \gamma \, \rho_{\mathrm{bud}} \, \mathrm{exceed}(x), \tag{14}$$

where $\mathrm{exceed}(x)$ is the budget-violation slack and ρ_{bud} is a large positive coefficient. In the code, this is exactly the role played by the penalized objective solved in `cvx-ll.gms`: the profit term appears with the opposite sign, and the lower-level terms are added through a penalty coefficient denoted by `gamma`.

The interpretation is standard. For small values of γ, the penalized problem prioritizes the upper-level profit term. As γ grows, the solution is pushed toward points that are increasingly consistent with lower-level optimality. This is the basic logic behind penalty methods for bilevel optimization, and it is the perspective that guides the algorithm used here.

Companion lower-level solve

The implementation also solves the pure lower-level problem

$$\min_x \Phi_{\mathrm{LL}}(x; \widehat{\alpha}, \widehat{\beta}), \tag{15}$$

using `cvx-sl.gms`. This companion solve is needed because the market-hyperparameter updates are based on comparing the market contributions implied by the penalized bilevel step and by the lower-level step. In practice, the algorithm alternates between these two problems:

- a penalized bilevel step, solved through `cvx-ll.gms`;
- a lower-level step, solved through `cvx-sl.gms`.

Hyperparameter update

After both solves are available, the market hyperparameters are updated by projected gradient-style steps. Denoting generically by g_α^{od} and g_β^{od} the search directions obtained from the two solves, the update reads

$$\widehat{\alpha}_+^{od} = \Pi_{[\underline{\alpha}, \overline{\alpha}]} \left(\widehat{\alpha}^{od} - \mu_\alpha g_\alpha^{od} \right), \tag{16}$$

$$\widehat{\beta}_+^{od} = \Pi_{[\underline{\beta}, \overline{\beta}]} \left(\widehat{\beta}^{od} - \mu_\beta g_\beta^{od} \right), \tag{17}$$

where Π denotes Euclidean projection onto the admissible interval and μ_α, μ_β are step sizes. In the current code, these projections are implemented by clipping `alfa_od` and `beta_od` after every update.

Algorithm 1: Penalty-based bilevel algorithm with outer sparsity updates

Input: Initial market weights $\widehat{\alpha}^{od}, \widehat{\beta}^{od}$, initial reference capacities $s^{\text{prev}}, sh^{\text{prev}}, a^{\text{prev}}$, step sizes μ_α, μ_β, penalty coefficient γ, numbers of outer and inner iterations.

```
for k = 1, ..., K_outer do
    build the reweighted deployment term from s^prev, sh^prev, a^prev;
    for t = 1, ..., K_inner do
        solve the penalized bilevel problem min_x P_γ(x; α̂, β̂);
        recover the penalized solution x^pen;
        update the logit-share bounds and related auxiliary quantities;
        solve the pure lower-level problem min_x Φ_LL(x; α̂, β̂);
        recover the lower-level solution x^ll;
        compute the projected update directions for α̂ and β̂ from x^pen and x^ll;
        update α̂, β̂ by projected gradient steps;
    end
    set (s^prev, sh^prev, a^prev) equal to the capacities of the latest penalized solution;
end
```

Output: Network capacities, hub structure, market shares, and learned market hyperparameters.

Resolution algorithm

The complete method contains two nested loops. The outer loop updates the MM reweighting coefficients used to approximate fixed costs. The inner loop solves the penalized bilevel problem and updates the market hyperparameters. A compact description is given in Algorithm 1.

This algorithm should be understood as a practical penalty method rather than as an exact reformulation of the bilevel problem. Its purpose is to produce network designs that are simultaneously sparse, behaviorally meaningful, and aligned with the true profit objective of the airline.

6 Computational Scheme

The previous section states the algorithmic structure at the model level. This section summarizes how that structure is instantiated in the current implementation.

The outer loop is a classical majorization-minimization procedure for sparse deployment costs. A fixed activation term $\kappa\,\mathbf{1}_{\{z>0\}}$ is approximated at iteration k through

$$\kappa\,\frac{z^{(k)}}{z^{(k-1)}+\varepsilon}, \tag{18}$$

which yields the reweighted penalties that are written to GAMS through the arrays `s_prev`, `sh_prev`, and `a_prev`. In the active airline implementation, the nonsmooth approximation is driven primarily by airport and hub capacities.

Inside that outer loop, `generate_data.py` runs an inner calibration loop for the market-wise coefficients. The workflow is:

- solve the penalized bilevel problem in `cvx-ll.gms`;
- recover capacities, flows, and market shares from the GAMS outputs;

- update auxiliary bounds for feasible demand capture;
- solve the pure lower-level problem in `cvx-sl.gms`;
- compute gradient-like corrections for `alfa_od` and `beta_od`;
- project the hyperparameters back to their admissible ranges;
- update the MM reference capacities for the next outer iteration.

The most relevant implementation detail is that two GAMS models are solved repeatedly. The first one, `cvx-ll.gms`, evaluates the penalty surrogate that mixes upper-level profit and lower-level optimality. The second one, `cvx-sl.gms`, evaluates the lower-level objective itself. Their outputs are then combined in Python to update the market hyperparameters and to refresh the reweighted fixed-cost approximation.

The resulting procedure is not intended as a globally exact solver. It is a structured successive-approximation method whose practical goal is to maintain the continuous scalability of the single-level model while improving the alignment between sparse deployment decisions and upper-level profit.

7 Conclusion

We have presented a continuous bilevel framework for airline hub-and-spoke design in which sparse deployment decisions, logit demand capture, and profit maximization are treated within a unified optimization model. The lower level determines capacities, flows, and market shares under a continuous approximation of fixed costs, while the upper level learns market-specific hyperparameters that steer the design toward more profitable market allocations.

The resulting model preserves the main practical advantages of continuous optimization. In particular, it remains scalable and directly compatible with rich passenger-choice terms. At the same time, the penalty-based bilevel strategy provides a concrete mechanism for coupling the learned market priorities with the true economic objective of the airline. For that reason, the framework is a promising alternative to exact mixed-integer bilevel formulations for large airline network design problems.

The next step is to deepen the numerical section: report the effect of the learned market hyperparameters on the resulting network topology, compare the penalty-based bilevel design against the underlying single-level model, and quantify how the outer sparsity loop affects both profitability and the number of hubs selected.